

Index

S.N	LAB Problems	Date	Signature
1	Write program to implement Monte Carlo's simulation	2082/02/15	
2	Write program to implement queuing system and calculate different queue parameters.	2082/02/15	
3	Write program to implement markov chain	2082/02/20	
4	Write program generate random number using linear congruential method	2082/02/20	
5	Write program generate random number using multiplicative congruential method.	2082/02/20	
6	Write program generate random number using mid square method.	2082/02/23	
7	Write program to implement KS-test	2082/02/23	
8	Write program to implement autocorrelation test	2082/02/24	
9	Write program to implement Chi-Square test	2082/02/24	
10	Write program to implement gap test	2082/02/24	
11	GPSS program: i. Joe's Barbershop (Event Driven) ii. Joe's Barbershop (Time Driven) iii. Manufacturing shop iv. Three Inspector Inspection v. Manufacturing shop model 4	2082/02/25	

Output of GAP Test:

```
PS D:\Arjun Mijar(109) Lab Reports\Simulation and Modeling> .\gapTest.exe
Enter total number of random numbers: 20
Enter the random numbers (between 0 and 1):
0.12 0.44 0.88 0.63 0.29 0.77 0.39 0.58 0.91 0.21 0.53 0.67 0.25 0.94 0.33 0.40 0.74 0.48 0.19 0.62
Enter interval [a, b] (e.g., 0.3 0.7): 0.3 0.7

Gap Length      Observed(O)      Expected(E)      (O-E)^2/E
0                3                4.40             0.45
1                3                2.64             0.05
2                3                1.58             1.27
3                0                0.95             0.95
4                0                0.57             0.57
5                0                0.34             0.34
6                0                0.21             0.21
7                0                0.12             0.12
8                0                0.07             0.07
9                0                0.04             0.04
10               0                0.17             0.17

Chi-square = 4.2361
Enter critical chi-square value (from table) for significance level and dof: 9.488
Result: ACCEPT null hypothesis (numbers are independent)
```

Output of Monte Carlo's Simulation:

```
PS D:\Arjun Mijar(109) Lab Reports\Simulation and Modeling> .\MonteCarloPi.exe
Estimated value of Pi using 10000 samples is: 3.146000
```

Output of Queuing System

```
PS D:\Arjun Mijar(109) Lab Reports\Simulation and Modeling> .\QueuingSystem.exe
Enter arrival rate (lambda): 2
Enter service rate (mu): 3

--- Queue Parameters ---
Server Utilization ( $\rho$ ): 0.667
Average number of customers in system (L): 2.000
Average number of customers in queue (Lq): 1.333
Average time a customer spends in system (W): 1.000 units
Average waiting time in queue (Wq): 0.667 units
```

Output of Markov Chain:

```
PS D:\Arjun Mijar(109) Lab Reports\Simulation and Modeling> .\markovChain.exe
Enter the transition matrix (2 x 2):
0.7 0.3
0.4 0.6
Enter initial state vector (2 values):
1 0
Enter number of steps: 3

Initial State:
1.0000 0.0000
After Step 1:
0.7000 0.3000
After Step 2:
0.6100 0.3900
After Step 3:
0.5830 0.4170
```

Output of Linear Congruential Method:

```
PS D:\Arjun Mijar(109) Lab Reports\Simulation and Modeling> .\RandomNumberLCM.exe
Enter the seed value: 1234
Enter the number of random numbers to generate: 5

Generated Random Numbers:
0.714308
0.207013
0.749542
0.346062
0.803439
```

Output of Multiplicative Congruential Method:

```
PS D:\Arjun Mijar(109) Lab Reports\Simulation and Modeling> .\randomNumMultiCM.exe
Enter the seed value: 54321
Enter the number of random numbers to generate: 5

Generated Random Numbers:
0.425136
0.264190
0.236707
0.333816
0.449103
```

Output of Mid-Square Method:

```
PS D:\Arjun Mijar(109) Lab Reports\Simulation and Modeling> .\randNoMidSquare.exe
Enter the seed value (example: 4 or 6 digit number): 1234
Enter the number of random numbers to generate: 5
Enter the number of middle digits to extract (e.g., 4): 4

Generated Random Numbers:
0.229000
0.446200
0.911300
0.055000
0.025300
```

Output of Kolmogorov-Smirnov (KS) Test:

```
PS D:\Arjun Mijar(109) Lab Reports\Simulation and Modeling> .\KSTest.exe
Enter the number of random numbers: 5
Enter the random numbers (between 0 and 1):
0.12
0.58
0.33
0.77
0.44

D+ = 0.230000
D- = 0.130000
D = 0.230000

Enter critical value for KS test (example: for n=5 at 5% level, D=0.565): 0.565

Result: ACCEPT null hypothesis (uniform distribution)
```

Output of Autocorrelation Test:

```
PS D:\Arjun Mijar(109) Lab Reports\Simulation and Modeling> .\autoCorrelationTest.exe
Enter the number of random numbers: 10
Enter the random numbers (between 0 and 1):
0.23
0.56
0.12
0.77
0.45
0.98
0.11
0.67
0.89
0.32
Enter the lag value (k): 2

Autocorrelation Coefficient (r_k) = 0.285725
The numbers are not independent (High autocorrelation).
```

Output of Chi-Square Test:

```
PS D:\Arjun Mijar(109) Lab Reports\Simulation and Modeling> .\chiSquareTest.exe
Enter the number of random numbers: 10
Enter the number of intervals (e.g., 5 or 10): 5
Enter the random numbers (between 0 and 1):
0.15
0.67
0.89
0.45
0.23
0.78
0.34
0.01
0.59
0.44

Interval      Observed(O)      Expected(E)      (O-E)^2/E
1             2                2.00            0.00
2             2                2.00            0.00
3             3                2.00            0.50
4             2                2.00            0.00
5             1                2.00            0.50

Chi-Square value = 1.0000
```

11. i) Joe's Barbershop (Event Driven)

```
GPSS World Simulation Report - Joe's Barbershop (event-driven) by Arjun Mijar.1.1

Wednesday, June 18, 2025 07:32:53

START TIME      END TIME  BLOCKS  FACILITIES  STORAGES
0.000           937.274    7        1           0

NAME            VALUE
JOE              10001.000
SEAT             10000.000

LABEL           LOC  BLOCK TYPE      ENTRY COUNT  CURRENT COUNT  RETRY
1      GENERATE      50          0        0
2      QUEUE        50          0        0
3      SEIZE        50          0        0
4      DEPART       50          0        0
5      ADVANCE      50          0        0
6      RELEASE      50          0        0
7      TERMINATE    50          0        0

FACILITY        ENTRIES  UTIL.   AVE. TIME AVAIL.  OWNER  PEND  INTER  RETRY  DELAY
JOE              50    0.796   14.922  1      0    0    0    0    0

QUEUE           MAX CONT.  ENTRY  ENTRY(0)  AVE.CONT.  AVE.TIME  AVE.(-0)  RETRY
SEAT             1    0    50    37    0.046    0.869    3.342    0

FEC XN  PRI      BDT      ASSEM  CURRENT  NEXT  PARAMETER  VALUE
51      0      944.962    51      0      1
```

ii) Joe's Barbershop (Time-Driven)

GPSS World Simulation Report - Joe's Barbershop (time-driven) by Arjun Mijar.13.1									
Thursday, June 19, 2025 19:38:01									
START TIME		END TIME		BLOCKS	FACILITIES	STORAGES			
0.000		8107.355		7	1	1			
NAME		VALUE							
JOB	10000.000								
SEAT	10001.000								
LABEL	LOC	BLOCK TYPE	ENTRY COUNT	CURRENT COUNT	RETRY				
	1	GENERATE	810	0	0				
	2	QUEUE	810	269	0				
	3	SEIZE	541	1	0				
	4	DEPART	540	0	0				
	5	ADVANCE	540	0	0				
	6	RELEASE	540	0	0				
	7	TERMINATE	540	0	0				
FACILITY	ENTRIES	UTIL.	AVE. TIME	AVAIL.	OWNER	PEND	INTER	RETRY	DELAY
JOB	541	0.999	14.967	1	541	0	0	0	269
QUEUE	MAX CONT.	ENTRY	ENTRY(0)	AVE.CONT.	AVE.TIME	AVE.(-0)	RETRY		
SEAT	270	270	810	1	135.078	1352.002	1353.674	0	
STORAGE	CAP.	REM.	MIN.	MAX.	ENTRIES	AVL.	AVE.C.	UTIL.	RETRY DELAY
JOB	1	1	0	0	0	1	0.000	0.000	0 0
CEC XN	PRI	M1	ASSEM	CURRENT	NEXT	PARAMETER	VALUE		
541	0	5410.000	541	3	4				
FEC XN	PRI	BDT	ASSEM	CURRENT	NEXT	PARAMETER	VALUE		
811	0	8110.000	811	0	1				

iii) Manufacturing shop

GPSS World Simulation Report - Manufacturing Shop (By Arjun Mijar).1.1									
Thursday, June 19, 2025 07:31:52									
START TIME		END TIME		BLOCKS	FACILITIES	STORAGES			
0.000		2504.589		9	1	0			
NAME				VALUE					
ACC				8.000					
INSPECTOR				10001.000					
REJ				9.000					
SHOP				10000.000					
LABEL	LOC	BLOCK TYPE	ENTRY COUNT	CURRENT COUNT	RETRY				
	1	GENERATE	500	0	0				
	2	QUEUE	500	0	0				
	3	SEIZE	500	0	0				
	4	DEPART	500	0	0				
	5	ADVANCE	500	0	0				
	6	RELEASE	500	0	0				
	7	TRANSFER	500	0	0				
ACC	8	TERMINATE	453	0	0				
REJ	9	TERMINATE	47	0	0				
FACILITY	ENTRIES	UTIL.	AVE. TIME	AVAIL.	OWNER	PEND	INTER	RETRY	DELAY
INSPECTOR	500	0.770	3.857	1	0	0	0	0	0
QUEUE	MAX CONT.	ENTRY	ENTRY(0)	AVE.CONT.	AVE.TIME	AVE.(-0)	RETRY		
SHOP	1	0	500	285	0.111	0.556	1.293	0	
FEC XN	PRI	BDT	ASSEM	CURRENT	NEXT	PARAMETER	VALUE		
501	0	2505.000	501	0	1				

iv) Three Inspector Inspection

GPSS World Simulation Report - 3_inspectorInspection by Arjun Mijar.2.1

Thursday, June 19, 2025 08:19:44

START TIME	END TIME	BLOCKS	FACILITIES	STORAGES
0.000	5009.809	9	0	1

NAME	VALUE
ACC	8.000
LINE	10001.000
REJ	9.000
REPO	10000.000

LABEL	LOC	BLOCK TYPE	ENTRY COUNT	CURRENT COUNT	RETRY
	1	GENERATE	1001	0	0
	2	QUEUE	1001	0	0
	3	ENTER	1001	0	0
	4	DEPART	1001	0	0
	5	ADVANCE	1001	1	0
	6	LEAVE	1000	0	0
	7	TRANSFER	1000	0	0
ACC	8	TERMINATE	887	0	0
REJ	9	TERMINATE	113	0	0

QUEUE	MAX CONT.	ENTRY	ENTRY(0)	AVE.CONT.	AVE.TIME	AVE.(-0)	RETRY
LINE	2	0	1001	753	0.105	0.525	2.118 0

STORAGE	CAP.	REM.	MIN.	MAX.	ENTRIES	AVL.	AVE.C.	UTIL.	RETRY	DELAY
REPO	3	2	0	3	1001	1	2.332	0.777	0	0

FEC XN	PRI	BDT	ASSEM	CURRENT	NEXT	PARAMETER	VALUE
1002	0	5010.000	1002	0	1		
1001	0	5024.074	1001	5	6		

v) Manufacturing shop model 4

GPSS World Simulation Report - manufacturingShopModel4 by Arjun Mijar.2.1

Thursday, June 19, 2025 19:56:16

START TIME	END TIME	BLOCKS	FACILITIES	STORAGES
0.000	5013.849	11	0	1

NAME	VALUE
ACC	10.000
REJ	11.000
SEAT	10002.000
STO	10000.000
TAB	10001.000

LABEL	LOC	BLOCK TYPE	ENTRY COUNT	CURRENT COUNT	RETRY
	1	GENERATE	1002	0	0
	2	QUEUE	1002	0	0
	3	ENTER	1002	0	0
	4	DEPART	1002	0	0
	5	MARK	1002	0	0
	6	ADVANCE	1002	2	0
	7	LEAVE	1000	0	0
	8	TABULATE	1000	0	0
	9	TRANSFER	1000	0	0
ACC	10	TERMINATE	907	0	0
REJ	11	TERMINATE	93	0	0

QUEUE	MAX CONT.	ENTRY	ENTRY(0)	AVE.CONT.	AVE.TIME	AVE.(-0)	RETRY
SEAT	1	0	1002	992	0.001	0.006	0.593 0

STORAGE	CAP.	REM.	MIN.	MAX.	ENTRIES	AVL.	AVE.C.	UTIL.	RETRY	DELAY
STO	4	2	0	4	1002	1	2.286	0.597	0	0

TABLE	MEAN	STD.DEV.	RANGE	RETRY FREQUENCY	CUM. %
TAB	11.947	8.195	-	0	
			-	5.000	109 10.90
			5.000 -	10.000	280 38.90
			10.000 -	15.000	287 67.60
			15.000 -	20.000	266 94.20
			20.000 -	25.000	58 100.00

FEC XN	PRI	BDT	ASSEM	CURRENT	NEXT	PARAMETER	VALUE
1003	0	5015.000	1003	0	1		
1000	0	5020.832	1000	6	7		
1002	0	5022.045	1002	6	7		